

What is Process Management?

Process management means monitoring, controlling, starting, stopping, and troubleshooting processes running on a server.

1) ps

```
ubuntu@ip-172-31-28-192:~$ ps
  PID TTY          TIME CMD
 1257 pts/1    00:00:00 bash
 1613 pts/1    00:00:00 ps
ubuntu@ip-172-31-28-192:~$ |
```

2) ps aux

```
ubuntu@ip-172-31-28-192:~$ ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root             1   0.1  1.4  22584  13752 ?        Ss   13:56   0:01 /sbin/init
root             2   0.0  0.0      0     0 ?        S    13:56   0:00 [kthreadd]
root             3   0.0  0.0      0     0 ?        S    13:56   0:00 [pool_workque
root             4   0.0  0.0      0     0 ?        I<   13:56   0:00 [kworker/R-rc
root             5   0.0  0.0      0     0 ?        I<   13:56   0:00 [kworker/R-sy
root             6   0.0  0.0      0     0 ?        I<   13:56   0:00 [kworker/R-kv
root             7   0.0  0.0      0     0 ?        I<   13:56   0:00 [kworker/R-sl
root             8   0.0  0.0      0     0 ?        I<   13:56   0:00 [kworker/R-ne
root             9   0.0  0.0      0     0 ?        I    13:56   0:00 [kworker/0:0-
root            10   0.0  0.0      0     0 ?        I<   13:56   0:00 [kworker/0:0H
root            12   0.0  0.0      0     0 ?        I    13:56   0:00 [kworker/u60:
root            13   0.0  0.0      0     0 ?        I<   13:56   0:00 [kworker/R-mm
root            14   0.0  0.0      0     0 ?        I    13:56   0:00 [rcu_tasks_ru
root            15   0.0  0.0      0     0 ?        I    13:56   0:00 [rcu_tasks_tr
root            16   0.0  0.0      0     0 ?        S    13:56   0:00 [ksoftirqd/0]
root            17   0.0  0.0      0     0 ?        I    13:56   0:00 [rcu_sched]
root            18   0.0  0.0      0     0 ?        S    13:56   0:00 [rcu_exp_par_
root            19   0.0  0.0      0     0 ?        S    13:56   0:00 [rcu_exp_gp_k
root            20   0.0  0.0      0     0 ?        S    13:56   0:00 [migration/0]
root            21   0.0  0.0      0     0 ?        S    13:56   0:00 [idle_inject/
root            22   0.0  0.0      0     0 ?        S    13:56   0:00 [cpuhp/0]
root            23   0.0  0.0      0     0 ?        S    13:56   0:00 [kdevtmpfs]
root            24   0.0  0.0      0     0 ?        I<   13:56   0:00 [kworker/R-in
```

3) htop - The htop command is an interactive, real-time system monitor, process viewer, and process manager for Unix-based system

4) top -

```
ubuntu@ip-172-31-28-192:~$ top
top - 14:15:41 up 19 min,  2 users,  load average: 0.00, 0.00, 0.00
Tasks: 111 total,  1 running, 110 sleeping,  0 stopped,  0 zombie
%Cpu(s):  0.3 us,  0.0 sy,  0.0 ni, 77.7 id,  0.0 wa,  0.0 hi,  0.0 si, 22.0 st
MiB Mem :  957.3 total,  297.3 free,  353.8 used,  466.6 buff/cache
MiB Swap:   0.0 total,   0.0 free,   0.0 used.  603.6 avail Mem
```

5)systemctl :

```
ubuntu@ip-172-31-28-192:~$ systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-02-10 14:09:50 UTC; 10min ago
     Docs: man:nginx(8)
```

Imp command

systemctl status docker

systemctl restart jenkins

systemctl enable nginx

systemctl stop mysql

Show all running processes (most used)

ps aux

a → all users

u → user-oriented format

x → include background/daemon processes

EXAMPLES

To see every process on the system using standard syntax:

ps -e

ps -ef

ps -eF

ps -ely

To see every process on the system using BSD syntax:

ps ax

ps axu

To print a process tree:

ps -ejH

ps axjf

To get info about threads:

ps -eLf

ps axms

To get security info:

ps -eo euser,ruser,suser,fuser,f,comm,label

ps axZ

ps -eM

To see every process running as root (real & effective ID) in user format:

ps -U root -u root u

To see every process with a user-defined format:

ps -eo pid,tid,class,rtprio,ni,pri,psr,pcpu,stat,wchan:14,comm

ps axo stat,euid,ruid,tty,tpgid,sess,pgrp,ppid,pid,pcpu,comm

ps -Ao pid,tt,user,fname,tmout,f,wchan

Print only the process IDs of syslogd:

ps -C syslogd -o pid=

Print only the name of PID 42:

File system

pwd - Print Working Directory

ls - List Directory Contents

cd - Change Directory

mkdir - Make Directories

rmdir - Remove Empty Directories

cp - Copy Files and Directories

mv - Move or Rename Files and Directories

cat - Concatenate and Display File Content

Networking troubleshooting –

ping

ip addr

dig

curl

ping -

```
ubuntu@ip-172-31-28-192:~$ ping google.com
PING google.com (142.250.177.78) 56(84) bytes of data.
64 bytes from pnorda-ao-in-f14.1e100.net (142.250.177.78): icmp_seq=1 ttl=117 time=8.12 ms
64 bytes from pnorda-ao-in-f14.1e100.net (142.250.177.78): icmp_seq=2 ttl=117 time=8.15 ms
64 bytes from pnorda-ao-in-f14.1e100.net (142.250.177.78): icmp_seq=3 ttl=117 time=8.37 ms
64 bytes from pnorda-ao-in-f14.1e100.net (142.250.177.78): icmp_seq=4 ttl=117 time=8.07 ms
64 bytes from pnorda-ao-in-f14.1e100.net (142.250.177.78): icmp_seq=5 ttl=117 time=7.74 ms
64 bytes from pnorda-ao-in-f14.1e100.net (142.250.177.78): icmp_seq=6 ttl=117 time=8.48 ms
```

```
ubuntu@ip-172-31-28-192:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enX0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc fq_codel state UP group default qlen 1000
    link/ether 06:98:f4:a0:16:1b brd ff:ff:ff:ff:ff:ff
    inet 172.31.28.192/20 metric 100 brd 172.31.31.255 scope global dynamic enX0
        valid_lft 3180sec preferred_lft 3180sec
    inet6 fe80::498:f4ff:fea0:161b/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@ip-172-31-28-192:~$
```

```
ubuntu@ip-172-31-28-192:~$ dig

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>>
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 8242
;; flags: qr rd ra; QUERY: 1, ANSWER: 13, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
; .                               IN      NS

;; ANSWER SECTION:
.           86359    IN      NS      h.root-servers.net.
.           86359    IN      NS      l.root-servers.net.
.           86359    IN      NS      j.root-servers.net.
.           86359    IN      NS      g.root-servers.net.
.           86359    IN      NS      b.root-servers.net.
.           86359    IN      NS      i.root-servers.net.
.           86359    IN      NS      m.root-servers.net.
.           86359    IN      NS      c.root-servers.net.
.           86359    IN      NS      a.root-servers.net.
.           86359    IN      NS      d.root-servers.net.
.           86359    IN      NS      k.root-servers.net.
.           86359    IN      NS      e.root-servers.net.
.           86359    IN      NS      f.root-servers.net.

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Tue Feb 10 14:34:08 UTC 2026
;; MSG SIZE rcvd: 239
```

Curl - is one of the most useful commands for **DevOps engineers**, **cloud engineers**, **Linux admins**, and **developers**.

It is used to **send requests to URLs** and interact with web servers, APIs, and services.

You can use it to:

- Test websites
- Test APIs
- Download files

- Check if servers are running
- Debug networking issues
- Send data (POST, PUT, DELETE)
- Check response headers
- Verify connectivity between services